

# Groundline v2

An AI brand-intelligence analyst that shows its sources

Reads everything said about Weave and its competitors · alerts on risk · writes cited briefs · a second model fact-checks every claim

# A3 collected the data. A4 thinks about it.

BEFORE / AFTER

## Assignment 3 collected the data. Assignment 4 thinks about it.

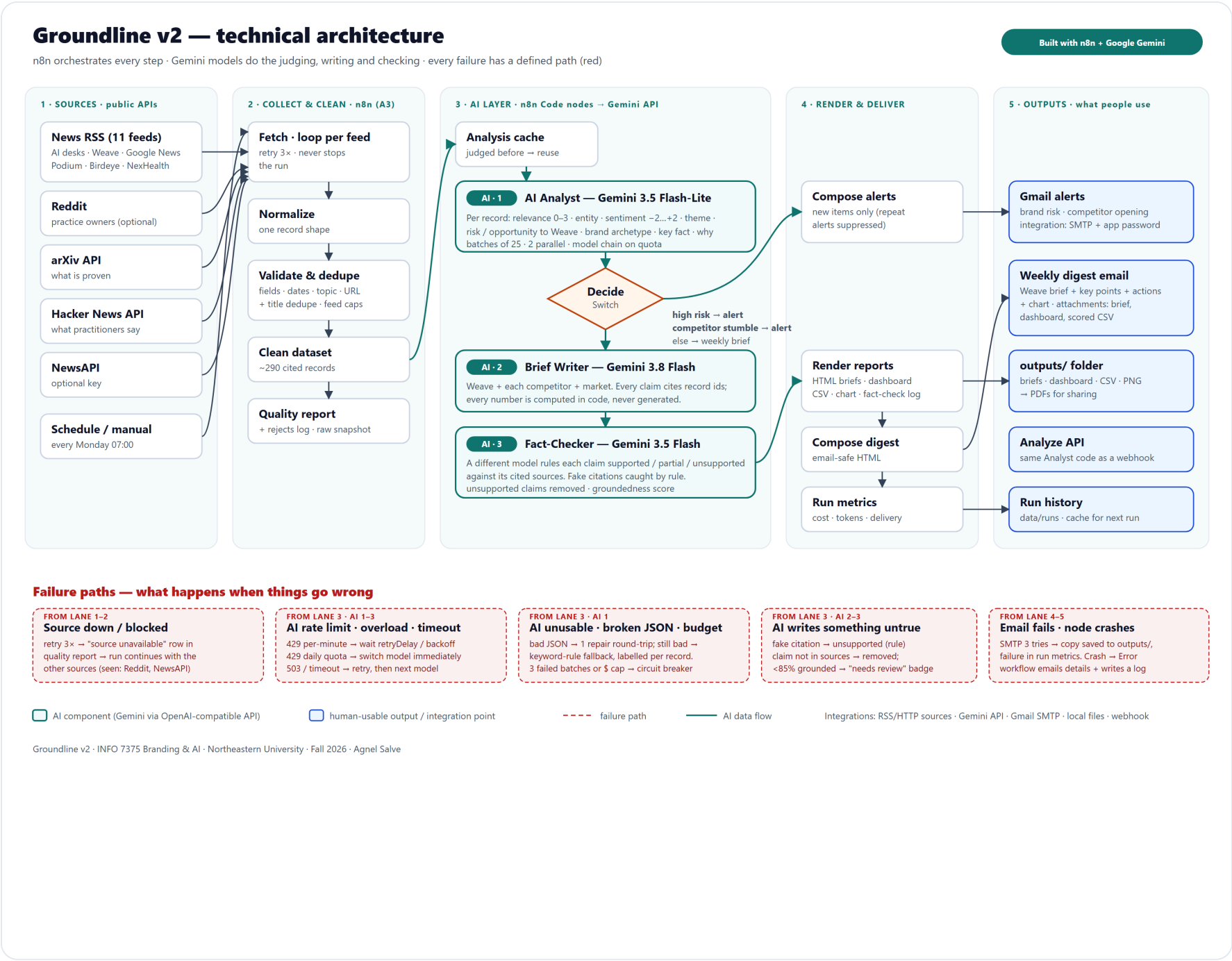
Same pipeline underneath — v2 adds the intelligence layer on top.

|                | A3 • GROUNDLINE DATA PIPELINE                        | A4 • GROUNDLINE V2 BRAND-INTELLIGENCE AGENT                                                                 |
|----------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| What it does   | Collects, validates and de-duplicates data           | Judges every record, decides, alerts, writes briefs, fact-checks them, delivers                             |
| AI             | None — keyword rules only                            | 3 Gemini models with separate jobs (judge · write · check), Muse Spark optional                             |
| Output         | CSV / JSON dataset + quality report                  | Gmail digest + risk & opportunity alerts, cited HTML/PDF briefs, dashboard, scored CSV, chart               |
| Brand focus    | 9% of records about Weave; no competitors            | Weave + Podium, Birdeye, NexHealth...: share of voice, sentiment, archetype                                 |
| Relevance      | Keyword match — an F1 race 'podium' counts as Podium | AI relevance check removed 13 off-topic items                                                               |
| Trust          | Every record cited                                   | Every claim cited, fact-checked by a second model, unsupported claims removed, groundedness score           |
| Error handling | Retry each source 3×                                 | + backoff, quota-aware model switching, JSON repair, rule fallback, circuit breaker, \$ cap, error workflow |
| Scale          | Not tested                                           | Tested 1 → 2,000 requests; breaking points found and one fixed                                              |
| Cost tracking  | —                                                    | Tokens, cost, retries and fallbacks recorded for every run                                                  |

- A3: 220 cited records, keyword rules, no output a person reads.
- A4: three Gemini models with separate jobs — **judge, write, check**.
- Outputs land in a real Gmail inbox: digest, risk alerts, opportunity alerts.
- Built for Weave's brand / data team (my Madison Intelligence-Agent project).

**Say:** “Last time I built the pipe. This time the pipe reads, decides and writes — and it has to prove every sentence.”

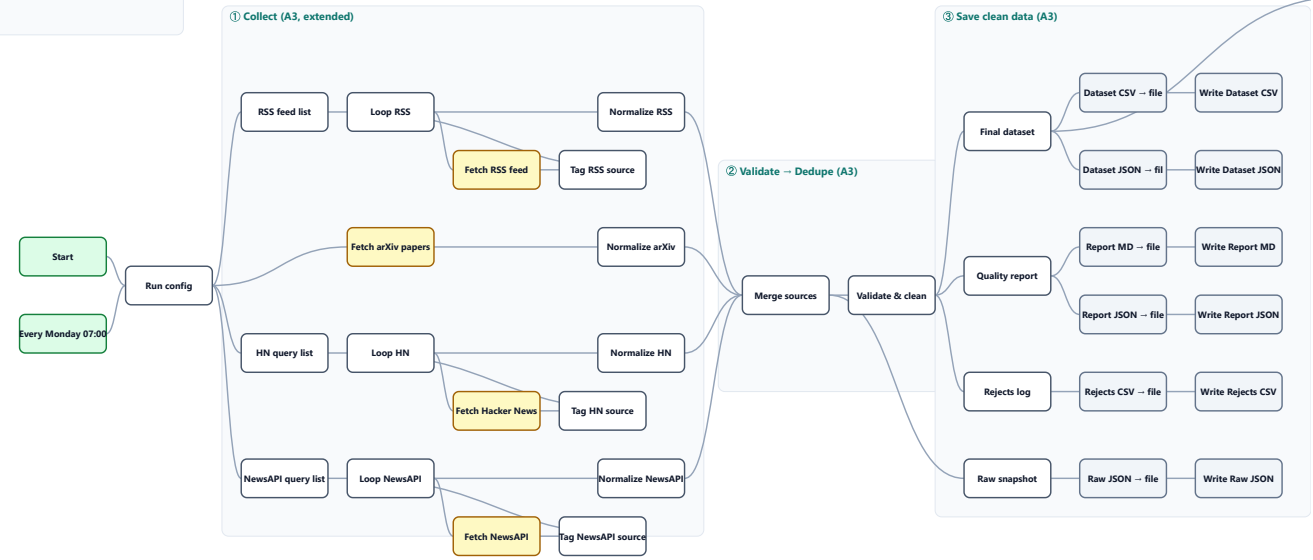
# Architecture: five lanes, every failure has a path



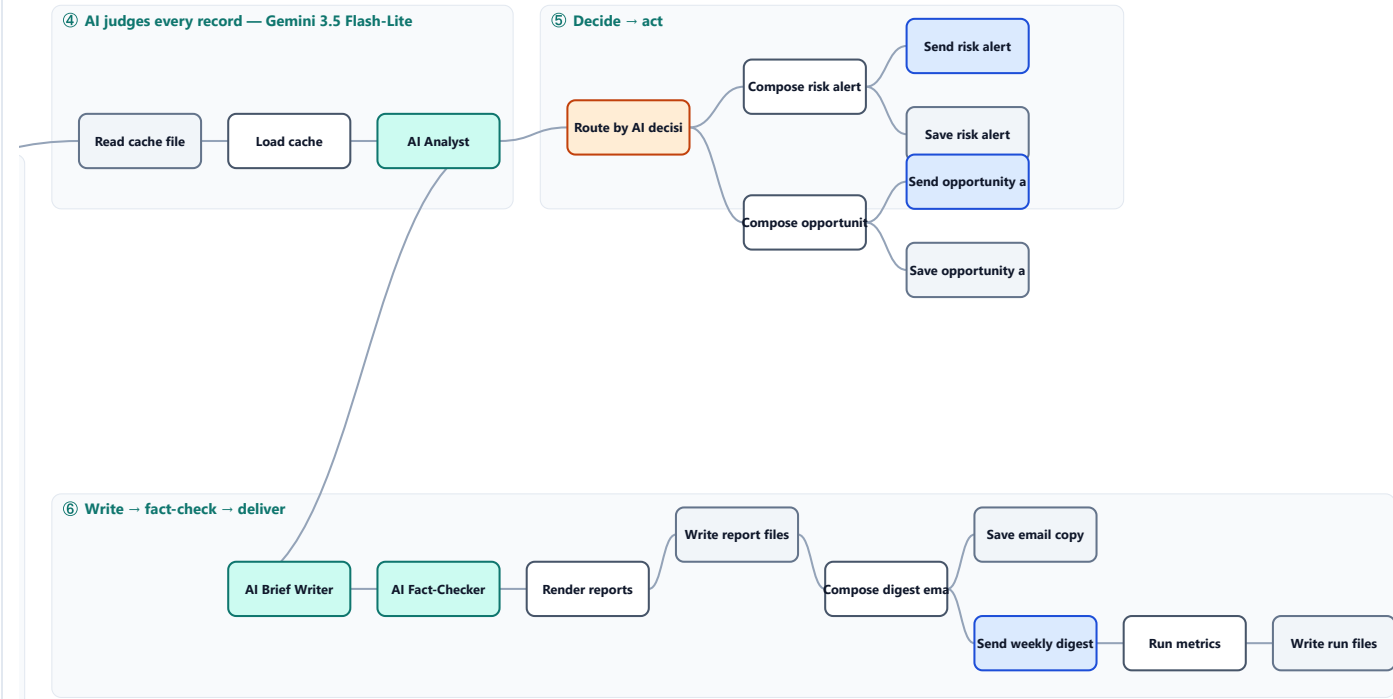
# 57 n8n nodes, one importable file

A3 · collect, validate, save

Groundline v2 — Brand Intelligence Agent



A4 · judge → decide → alert · write → fact-check → deliver



- Teal: the three AI Code nodes · orange: the Switch decision · blue: Gmail · yellow: source fetches.
- Two more workflows: an **Error handler** and the **Analyze API** webhook used for load tests.
- Code lives in workflow/src + workflow/lib; one build script makes the JSON.

**Say:** “Everything is in workflow\_v2.json — import it and press Execute.” (Canvas drawn from the real workflow file.)

# Gemini judges every record — and throws out the noise

OUTPUTS/DATA/2026-10-09\_SCORED\_RECORDS.CSV • 294 ROWS • OPENS IN EXCEL

Every record with the AI's judgement

| ENTITY | RELEVANCE | SENTIMENT | THEME            | RISK TO WEAVE | OPPORTUNITY FOR WEAVE | ARCHETYPE | ROUTE       | TITLE                                                                  |
|--------|-----------|-----------|------------------|---------------|-----------------------|-----------|-------------|------------------------------------------------------------------------|
| Weave  | 3         | 0         | financials       | low           | none                  |           | digest      | Form 4 Weave Communications Inc For: 8 October By Investing.com        |
| Weave  | 3         | 0         | financials       | low           | none                  |           | digest      | A holdings correction at Weave Communications (WEAV) follows Tyler New |
| Weave  | 3         | -1        | legal_regulatory | high          | none                  | Outlaw    | urgent_risk | WEAV Stock Alert: Halper Sadeh LLC is Investigating Whether Weave Comm |
| Weave  | 3         | -1        | financials       | low           | none                  |           | digest      | Weave Communications Q2 2026 Earnings: Modest EPS Beat Fails to Spark  |
| Weave  | 3         | 0         | financials       | low           | none                  |           | digest      | Weave (NASDAQ: WEAV) holder lines up 32K-share NYSE sale               |
| Weave  | 3         | -1        | financials       | medium        | none                  |           | digest      | Raymond James Downgrades Weave Communications to Market Perform        |
| Weave  | 3         | -1        | financials       | medium        | none                  |           | digest      | Raymond James Downgrades Weave Communications to Market Perform From O |
| Weave  | 3         | -1        | financials       | medium        | none                  |           | digest      | Raymond James Downgrades Weave Communications(WEAV.US) to Hold Rating, |
| Weave  | 3         | 0         | m_and_a          | low           | none                  |           | digest      | Raymond James cuts Weave Communications stock rating on buyout deal    |
| Weave  | 3         | 2         | m_and_a          | low           | high                  | Ruler     | opportunity | Weave Communications to Be Acquired by Francisco Partners for \$650M   |
| Weave  | 3         | -1        | legal_regulatory | medium        | none                  |           | digest      | Weave Communications Investor Alert: Kahn Swick & Foti, LLC Investigat |
| Weave  | 3         | 0         | m_and_a          | low           | none                  |           | digest      | Weave Communications (WEAV) Stock Fair Value Falls After Analysts Alig |
| Weave  | 3         | -1        | financials       | medium        | none                  |           | digest      | Loop Capital Downgrades Weave Communications to Hold From Buy, Adjusts |
| Weave  | 3         | -1        | m_and_a          | low           | none                  |           | digest      | Loop Capital downgrades Weave Communications stock on acquisition news |

- 294 records: relevance 0–3, entity, sentiment –2...+2, risk, opportunity, archetype, key fact, reason.
- **14 off-topic items removed** — F1 and endurance-racing “podium” stories that A3's keyword rules kept.
- Cache: second run judged 289 of 294 from cache — step time 41 s → 4 s.
- Exported as a spreadsheet anyone can open.

**Say:** “The AI isn't summarising — it's making 294 small judgements, each with a stated reason.”

# High risk → an email the same run

GROUNDLINE • BRAND INTELLIGENCE

## Brand risk: 1 new item need attention

Groundline flagged these items as a **risk to Weave's brand**. Each one links to its source so you can verify before acting.

Weave • Business Wire • 2026-09-19 • legal regulatory

### WEAV Stock Alert: Halper Sadeh LLC is Investigating Whether Weave Communications, Inc. is Obtaining a Fair Price for its Shareholders

Risk: high   Opportunity: none   Negative   reported

**What happened:** Halper Sadeh LLC is investigating whether Weave Communications, Inc. is obtaining a fair price for its shareholders.

**Why it was flagged:** Legal investigations pose severe risks to Weave's stock value and shareholder confidence.

Record GL-6e17f34a • judged by ai:gemin

Decision rule: relevance ≥ 2 and risk\_to\_weave = high (or a very negative Weave item rated medium).  
Run 20261009T045254Z • alerts are sent once per record (repeats are suppressed).

- Switch rule: relevance ≥ 2 and risk\_to\_weave = high → **risk alert**.
- Real catch: a law firm investigating the \$650M take-private price.
- Competitor stumble or market shift → **opportunity alert** (e.g. ServiceTitan dropping Podium; AI receptionists failing patients).
- Already-alerted items are remembered — no repeat pages.

**Say:** “This is the decision step: it doesn't just label things, it routes them and acts.”

# Every claim cited — and checked by another model

GROUNDLINE BRAND INTELLIGENCE • WEEK OF 2026-10-09

## Weave’s \$650M acquisition by Francisco Partners dominates coverage amid ongoing shareholder legal scrutiny

Weave brand brief • 36 items from 21 sources • 2026-07-13 to 2026-10-09

Needs review • 83% grounded

Weave Communications is transitioning to private ownership following a \$650M acquisition agreement. While the market responded positively to the deal, the brand faces legal investigations regarding the sale process and analyst downgrades.

|                     |                                   |                                                  |                                   |
|---------------------|-----------------------------------|--------------------------------------------------|-----------------------------------|
| ITEMS TRACKED<br>36 | AVG. SENTIMENT (−2...+2)<br>+0.44 | POSITIVE / NEUTRAL /<br>NEGATIVE<br>15 / 11 / 10 | RISKS • OPENINGS FLAGGED<br>8 • 3 |
|---------------------|-----------------------------------|--------------------------------------------------|-----------------------------------|

### What happened

- Weave Communications agreed to be acquired by Francisco Partners for \$650M, a move intended to take the company private.  
df38bbbc bded1b74
- The acquisition announcement triggered a surge in stock price and positive market attention. 5c9ff875 a59114d5
- Weave reported record location growth during its Q2 2026 earnings call prior to the acquisition news. b17a0a67

### How the brand is perceived

- Market analysts reacted to the acquisition with downgrades, shifting ratings to 'Hold' or 'Market Perform' due to the deal's constraints. 9cef8c1b 40e2d1d8 5779e31b
- The company is increasingly associated with legal and regulatory scrutiny regarding the fairness of the acquisition price. 6e17f34a d1464026

- Brief Writer gets computed STATS + evidence; it must cite record ids for every claim.
- Fact-Checker (different model) rules supported / partial / unsupported.
- Fabricated citations are caught by rule; unsupported claims are **removed** and listed.
- Today: 12 briefs checked, groundedness 54–97%; below 85% = “needs review” badge.

**Say:** “If the model makes something up, it doesn't reach the inbox — and the brief tells you how much of it was verified.”

# What lands in the inbox

GROUNDLINE • BRAND INTELLIGENCE

## Weave brand brief • 2026-10-09

needs review • 83% grounded

36 items • avg sentiment +0.44

12 alerts raised

### Weave's \$650M acquisition by Francisco Partners dominates coverage amid ongoing shareholder legal scrutiny

Weave Communications is transitioning to private ownership following a \$650M acquisition agreement. While the market responded positively to the deal, the brand faces legal investigations regarding the sale process and analyst downgrades.

KEY POINTS

- **What happened:** Weave Communications agreed to be acquired by Francisco Partners for \$650M, a move intended to take the company private.
- **How the brand is perceived:** Market analysts reacted to the acquisition with downgrades, shifting ratings to 'Hold' or 'Market Perform' due to the deal's constraints.
- **Risks to watch:** Multiple law firms are investigating whether Weave obtained a fair price for its shareholders in the proposed sale.

- Weekly digest: headline, key points, 3 actions, chart, competitor one-liners.
- Attached: full brief, dashboard, 294-row spreadsheet.
- 6 real emails delivered today — Gmail message ids in data/runs/.
- Copies of every email are also saved to outputs/.

**Say:** "A non-technical marketer gets this on Monday morning — no n8n, no JSON."

# When things go wrong, it says so

OUTPUTS/RUN\_SUMMARY.MD • MONITORING RECORD OF THE RUN

## Groundline run summary

Run 20261009T052422Z • 85.7s • 294 records judged • AI cost **\$0.0403**

| STEP                                        | SECONDS                      |               |              |
|---------------------------------------------|------------------------------|---------------|--------------|
| analyst                                     | 1.6                          |               |              |
| writer                                      | 17.3                         |               |              |
| checker                                     | 43.8                         |               |              |
| AI USAGE                                    |                              | VALUE         |              |
| Calls                                       |                              | 19            |              |
| Retries / JSON repairs / provider fallbacks |                              | 0 / 2 / 8     |              |
| Tokens in / out                             |                              | 29083 / 11223 |              |
| Records from cache                          |                              | 292           |              |
| Records judged by AI / by keyword fallback  |                              | 2 / 0         |              |
| Circuit breaker opened                      |                              | no            |              |
| BRIEF                                       | WRITTEN BY                   | GROUNDEDNESS  | STATUS       |
| weave                                       | gemini:gemini-3.1-flash-lite | 83%           | needs_review |
| podium                                      | gemini:gemini-3.1-flash-lite | 77%           | needs_review |
| birdeye                                     | gemini:gemini-3.1-flash-lite | 94%           | verified     |
| market                                      | gemini:gemini-3.1-flash-lite | 77%           | needs_review |
| EMAIL                                       | STATUS                       |               |              |
| weekly_digest                               | sent                         |               |              |
| risk_alert                                  | nothingtosend                |               |              |
| opportunity_alert                           | nothingtosend                |               |              |

Routes: digest 269 • discard 13 • opportunity 11 • urgent\_risk 1

### Errors handled

- brief:birdeye gemini-3.8-flash quota: Gemini 3.8-flash 429: {"code":429,"message":"You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: <https://ai.google.dev/gemini-api/docs/rate-limits>. To monitor your curren
- brief:podium gemini-3.8-flash quota: Gemini 3.8-flash 429: {"code":429,"message":"You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: <https://ai.google.dev/gemini-api/docs/rate-limits>. To monitor your curren
- brief:weave gemini-3.8-flash quota: Gemini 3.8-flash 429: {"code":429,"message":"You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: <https://ai.google.dev/gemini-api/docs/rate-limits>. To monitor your curren
- brief:birdeye gemini-3.5-flash quota: Gemini 3.5-flash 429: {"code":429,"message":"You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: <https://ai.google.dev/gemini-api/docs/rate-limits>. To monitor your curren
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- brief:podium gemini-3.5-flash quota: Gemini 3.5-flash 429: {"code":429,"message":"You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: <https://ai.google.dev/gemini-api/docs/rate-limits>. To monitor your curren
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- check:weave gemini-3.5-flash quota: Gemini 3.5-flash 429: {"code":429,"message":"You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: <https://ai.google.dev/gemini-api/docs/rate-limits>. To monitor your curren

- Real today: Gemini 3.8 Flash **503 overloaded** and **daily quota** exhausted → automatic switch to the next model.
- Broken JSON → one repair round-trip → keyword-rule fallback, labelled per record.
- Circuit breaker, \$1/run budget cap, error workflow email + log.
- No AI key at all → the run still finishes, briefs say “AI offline”.

**Say:** “I didn't simulate these errors — the free tier produced them, and the log shows each one handled.”

# Real load tests: what broke, and the fix

SCALE TESTING • REAL RUNS, 2026-10-09

## What happens at 10× and 100× the volume

| MODE           | REQUESTS | WALL   | P50   | P95    | SUCCESS | RETRIES | HTTP ERRORS | COST    | PEAK N8N |
|----------------|----------|--------|-------|--------|---------|---------|-------------|---------|----------|
| AI, before fix | 1        | 14.38s | 14.4s | 14.4s  | 0%      | 3       | —           | \$0.000 | 0.4 GB   |
| AI, before fix | 10       | 1.43s  | 1.4s  | 1.4s   | 100%    | 0       | —           | \$0.004 | 0.4 GB   |
| AI, before fix | 50       | 174.9s | 61.4s | 174.5s | 70%     | 36      | 500×15      | \$0.014 | 0.9 GB   |
| AI, after fix  | 50       | 79.03s | 14.5s | 78.7s  | 100%    | 61      | —           | \$0.019 | 1.0 GB   |
| AI, after fix  | 100      | 156.6s | 34.5s | 156.0s | 100%    | 165     | —           | \$0.038 | 1.3 GB   |
| AI, batch ×10  | 5 ×10    | 7.8s   | 3.6s  | 7.8s   | 100%    | 0       | —           | \$0.015 | 0.4 GB   |
| n8n only       | 1        | 0.11s  | 0.1s  | 0.1s   | 100%    | 0       | —           | \$0.000 | 0.4 GB   |
| n8n only       | 10       | 0.45s  | 0.4s  | 0.4s   | 100%    | 0       | —           | \$0.000 | 0.4 GB   |
| n8n only       | 50       | 1.9s   | 1.3s  | 1.9s   | 100%    | 0       | —           | \$0.000 | 0.7 GB   |
| n8n only       | 100      | 3.52s  | 2.3s  | 3.4s   | 100%    | 0       | —           | \$0.000 | 1.4 GB   |
| n8n only       | 250      | 10.36s | 5.6s  | 9.9s   | 100%    | 0       | —           | \$0.000 | 2.9 GB   |
| n8n only       | 500      | 17.62s | 11.4s | 17.0s  | 100%    | 0       | —           | \$0.000 | 5.5 GB   |
| n8n only       | 1000     | 39.34s | 24.6s | 37.7s  | 100%    | 0       | —           | \$0.000 | 10.2 GB  |
| n8n only       | 2000     | 85.03s | 49.7s | 83.8s  | 95.6%   | 0       | 500×88      | \$0.000 | 20.6 GB  |

Breaks first: n8n's Code runner

10 tasks at a time by default; a task waiting >60 s is dropped → 15 × HTTP 500 at 50 concurrent AI requests.  
Fix: runner concurrency 50, queue timeout 300 s → 0 errors at 50 and 100.

Then: Gemini free-tier quota

Per-minute limits absorbed by 61–165 retries; daily quota (~20 req/model on Flash) → automatic model switch. Pro models: 0 free quota.

Then: memory

~10 MB per in-flight request; 2,000 burst → 21 GB and 4.4% failures. Platform ceiling here ≈ 1,000 burst / 25 req/s.

Production

Weekly pipeline: 24/7-ready today. High-volume API: paid Gemini tier + n8n queue mode + batching (10× faster per record). ≈ \$9–11/month at 1,000 requests/day.

- 50 concurrent AI requests → 15 × HTTP 500: n8n runs 10 Code tasks at a time.
- Fix: runner concurrency 50, queue timeout 300 s → 0 errors at 50 and 100.
- n8n alone: 1,000-request burst 100%; 2,000 → 4.4% fail, 21 GB.
- ≈ \$0.038 per 100 requests → ≈ \$9–11/month at 1,000/day.

Say: “The first thing to break wasn't the AI — it was n8n's code runner. Finding that is the point of testing.”

# Business value and what's next

GROUNDLINE V2 • EXECUTIVE SUMMARY • INFO 7375 ASSIGNMENT 4

## An AI brand-intelligence analyst that shows its sources

Every agent answer, traceable to a source. Agnel Salve · Northeastern University · Fall 2026

Built with n8n + Google Gemini

### Problem

Weave's brand and data teams track reputation and competitors by hand across news, analyst notes, legal notices and practitioner chatter — slow, and easy to miss the story that matters. AI summaries could do it faster, but nobody can trust them, because nothing ties a claim back to a source.

### Solution approach

- **Collect** ~500 items a run from 17 public feeds — Weave, 5 competitors, AI-market news, research, Reddit, Hacker News (A3 pipeline, extended).
- **Judge** every item with Gemini 3.5 Flash-Lite: relevance, sentiment, risk, opportunity, brand archetype, key fact.
- **Decide & act:** high risk → instant Gmail alert; competitor stumble → opportunity alert; repeats suppressed.
- **Write** cited briefs with Gemini 3.8 Flash — numbers computed in code, never generated.
- **Fact-check** every claim with a *different* model; unsupported claims are removed and the brief is scored.
- **Survive failure:** retries, model chain on quota, JSON repair, circuit breaker, budget cap, error workflow.

### Business value

- Replaces an estimated **4–6 analyst hours a week** of reading, writing and source-checking for about **\$0.16 of AI per run**.
- Risks reach the inbox the run they appear — e.g. the shareholder-investigation alert on the \$650M take-private.
- Every sentence links to its source and carries a groundedness score, so the team can act on it.
- At 1,000 API requests/day: ≈ **\$9–11 a month** in AI (measured per-request cost).

### Current performance (measured 20261009)

**294**

RECORDS JUDGED PER RUN

**301s → 86s**

COLD → CACHED RUN

**\$0.16**

AI COST PER RUN AT LIST PRICE (\$0 ON FREE TIER)

**87%**

MEDIAN GROUNDEDNESS, 12 BRIEFS FACT-CHECKED (54–97%)

**13**

NOISE ITEMS THE AI FILTERED OUT

**6**

REAL EMAILS DELIVERED TO GMAIL (3 OF 3 TYPES)

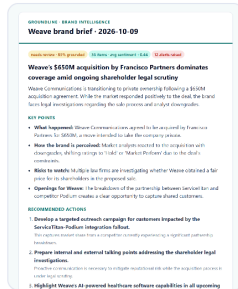
**100%**

SUCCESS AT 100 CONCURRENT AI REQUESTS

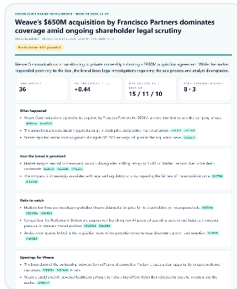
**1,000**

REQUEST BURST N8N HANDLES (100%)

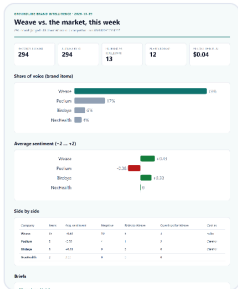
### Sample outputs



Weekly digest in Gmail



Cited Weave brief (HTML/PDF)



Weave vs. competitors dashboard

All numbers from real runs on 2026-10-09 — see `scale_test_results.md` and `data/runs/`. Costs at paid Gemini list prices; the free tier billed \$0. "Analyst hours" is an estimate.

- Replaces an estimated 4–6 analyst hours a week for ≈ \$0.16 of AI per run.
- Every sentence traceable to a source → a team can act on it.
- Next: paid Gemini tier + n8n queue mode for the API; route fix (entity ≠ Weave for opportunities).
- Then: feed the cited records into Madison's Intelligence Agent.

**Say:** "That's Groundline: cheaper than an analyst's afternoon, and it shows its work."

# Backup plan

## Before presenting

- Run `node scripts/check-setup.mjs` — confirms each Gemini model answers.
- Run once the night before; the cache makes the live run ~1–2 min.
- Keep `outputs/` open in a browser tab: briefs, dashboard, alerts.
- Keep this PDF and the Figma board open.

## If something breaks live

- **Gemini quota / 503** — that is the demo: show the model switch in `outputs/run_summary.md`.
- **No internet** — show yesterday's real outputs and the Gmail inbox screenshots.
- **n8n won't start** — open `workflow_v2.json` in the repo and walk this PDF.
- **Email delayed** — open the saved copy in `outputs/emails/`.

**Key points to land:** (1) three models with separate jobs, (2) every claim cited and independently checked, (3) real decisions → real emails, (4) honest scale limits with a fix, (5)  $\approx$  \$0.16 a run.